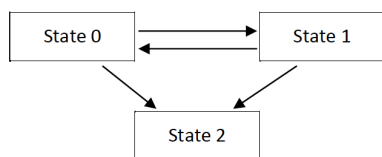


LTAM

8.11.

A continuous Markov process is modeled by the following multiple state diagram:



You are given the following constant transition intensities:

(i) $\mu^{01} = 0.08$

(ii) $\mu^{02} = 0.04$

(iii) $\mu^{10} = 0.10$

(iv) $\mu^{12} = 0.05$



For a person in State 1, calculate the probability that the person will continuously remain in State 1 for the next 15 years.

→ (A) 0.032

(B) 0.105

(C) 0.151

(D) 0.250

(E) 0.350

$${}_{15}P_x^{\pi} = e^{-\int_0^{15} (0.10 + 0.05) ds} = e^{-0.15(15)}$$

$$e^{-(0.10 + 0.05 + 0.08) \cdot 15} = 0.032$$

8.4.1 Kolmogorov's Forward Equations

Let i and j be any two, not necessarily distinct, states in a multiple state model which has $(n+1)$ states.

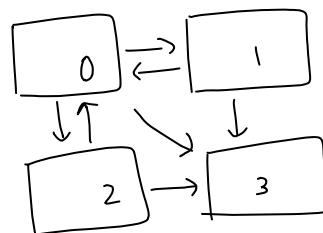
$$\frac{d}{dt} {}_tP_x^{ij} = \sum_{\substack{k=0 \\ k \neq j}}^n \left(\underbrace{{}_tP_x^{ik} \mu_{x+t}^{kj}}_{\text{Movement into state } j} - \underbrace{{}_tP_x^{ij} \mu_{x+t}^{jk}}_{\text{Movement out of state } j} \right)$$

↑
Change
in this
probability
with respect
to time

Boundary conditions

$$i=j \quad {}_0P_x^{ij} = 1$$

$$i \neq j \quad {}_0P_x^{ij} = 0$$

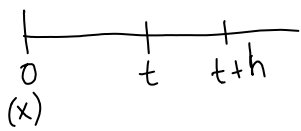


$$\frac{d}{dt} {}_tP_x^{10}$$

This gives us a set of equations known as Kolmogorov's forward equations.

Euler's method to approximate probabilities

h : step size in years



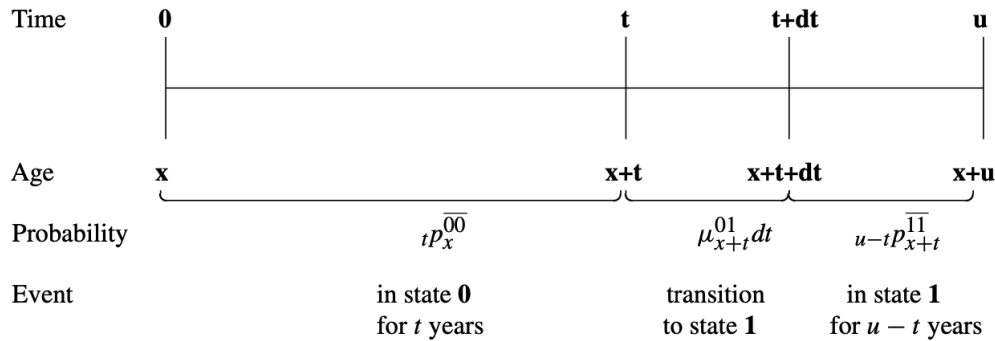
$${}_{t+h}P_x^{ij} = {}_tP_x^{ij} + h \sum_{\substack{k=0 \\ k \neq j}}^n ({}_tP_x^{ik} \mu_{x+t}^{kj} - {}_tP_x^{ij} \mu_{x+t}^{jk})$$

In one step size, movement is limited to one possible transition.

For any state i in a multiple state model,

$${}_t p_x^{\bar{ii}} = \exp \left\{ - \int_0^t \sum_{j=0, j \neq i}^n \mu_{x+s}^{ij} ds \right\}.$$

LTAM ${}_t p_x = e^{-\int_0^t \mu_{x+s} ds}$
 Constant μ ${}_t p_x = e^{-\mu t}$



$${}_u p_x^{01} = \int_0^u {}_t p_x^{\bar{00}} \mu_{x+t}^{01} {}_{u-t} p_{x+t}^{\bar{11}} dt.$$

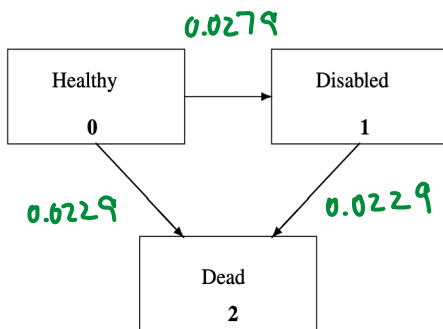


Figure 8.3 The permanent disability model.

Handwritten calculation of transition probabilities:

$$\begin{pmatrix} {}_{10}p_{60}^{00} & {}_{10}p_{60}^{01} & {}_{10}p_{60}^{02} \\ {}_{10}p_{60}^{10} & {}_{10}p_{60}^{11} & {}_{10}p_{60}^{12} \\ {}_{10}p_{60}^{20} & {}_{10}p_{60}^{21} & {}_{10}p_{60}^{22} \end{pmatrix} = \begin{pmatrix} 0.6017 & 0.1936 & \underline{0.2047} \\ 0 & 0.7953 & 0.2047 \\ 0 & 0 & 1 \end{pmatrix}$$

Note: ${}_{10}p_{60}^{02}$ is highlighted with a blue arrow pointing to 0.2047.

Example 8.4 Consider the permanent disability model illustrated in Figure 8.3.

(a) Suppose the transition intensities for this model are all constants, as follows

$$\mu_x^{01} = 0.0279, \quad \mu_x^{02} = 0.0229, \quad \mu_x^{12} = \mu_x^{02}.$$

Calculate ${}_{10}p_{60}^{00}$ and ${}_{10}p_{60}^{01}$.

Handwritten calculations:

$${}_{10}p_{60}^{00} = {}_{10}p_{60}^{\bar{00}} = e^{-\int_0^{10} (0.0279 + 0.0229) dt} = e^{-0.0508(10)} = \underline{0.6017}$$

$${}_{10}p_{60}^{01} = \int_0^{10} e^{-(0.0279 + 0.0229)t} \cdot (0.0279) \cdot e^{-0.0229(10-t)} dt$$

$$= \int_0^{10} e^{-0.0508t} \cdot (0.0279) \cdot e^{-0.0229(10-t)} \cdot e^{0.0229t} dt$$

$$= (0.0279) e^{-0.0229(10)} \int_0^{10} e^{-0.0279t} dt = \underline{0.1936}$$

$${}_{10}P_{60}^{11} = {}_{10}P_{60}^{\pi} = e^{-0.0229(10)} = \underline{0.7953}$$

$${}_{10}P_{60}^{12} = \int_0^{10} e^{-0.0229t} (0.0229) \cdot 1 dt = \underline{0.2047}$$

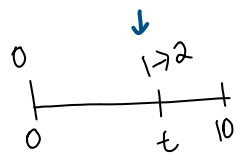
$$= 0.0229 \left. \frac{e^{-0.0229t}}{-0.0229} \right|_0^{10}$$

$${}_{10}P_{60}^{02} \quad \text{0} \rightarrow \text{2} \quad \int_0^{10} e^{-(0.0279 + 0.0229)t} \cdot (0.0229)(1) dt = \underline{0.1795}$$

Two different methods to solve

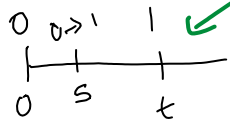
① $\int_0^{10} \frac{e^{-0.0508t}}{e^{-0.0229t}} \cdot (0.0279) (1 - e^{-0.0229(10-t)}) dt = \underline{0.0251}$ ←

negation not staying in state 1 = moving to state 2



② $\int_0^{10} t P_x^{01} \mu_{x+t}^{12} dt = \int_0^{10} (e^{-0.0229t} - e^{-0.0508t}) (0.0229) dt$

Need a function for $t P_x^{01}$



$$t P_x^{01} = \int_0^t e^{-0.0508s} \cdot (0.0279) e^{-0.0229(t-s)} ds$$

$$= 0.0279 e^{-0.0229t} \int_0^t e^{-0.0279s} ds$$

$$= \cancel{0.0279} e^{-0.0229t} \left[\frac{e^{-0.0279s}}{\cancel{0.0279}} \right]_0^t$$

$$= e^{-0.0229t} [1 - e^{-0.0279t}]$$

$$\star t P_x^{01} = e^{-0.0229t} - e^{-0.0508t} \star$$

$$\begin{aligned}
 {}_tP_x^{01} &= \int_0^{10} (e^{-0.0229t} - e^{-0.0508t}) (0.0229) dt \\
 &= (0.0229) \left[\frac{e^{-0.0229t}}{-0.0229} - \frac{e^{-0.0508t}}{-0.0508} \right]_0^{10} = 0.0251
 \end{aligned}$$

Andrey Nikolaevich Kolmogorov

• **Paul M.B. Vitanyi**, *CWI and Computer Science, University of Amsterdam, The Netherlands*



Figure 1: Kolmogorov in the 1940s

Andrei Nikolaevich Kolmogorov (Russian Андрей Никола́евич Колмогоров), born 25 April 1903 in Tambov, Russia, died 20 October 1987 in Moscow. He was perhaps the foremost contemporary Soviet mathematician and counts as one of the greatest mathematicians of the twentieth century. His many creative and fundamental contributions to a vast variety of mathematical fields are so wide-ranging that I cannot even attempt to treat them either completely or in any detail.

For now let me mention a non-exhaustive list of areas he enriched by his fundamental research: The theory of trigonometric series, measure theory, set theory, the theory of integration, constructive **logic** (intuitionism), topology, approximation theory, probability theory, the theory of random processes, information theory, mathematical statistics, **dynamical systems**, automata theory, theory of **algorithms**, mathematical **linguistics**, **turbulence** theory, **celestial mechanics**, differential equations, Hilbert's 13th problem, ballistics, and applications of mathematics to problems of biology, geology, and the crystallization of metals. In over 300 research papers, textbooks and monographs, Kolmogorov covered almost every area of mathematics except number theory. In all of these areas even his short contributions did not just study an isolated question, but in contrast exposed fundamental insights and deep relations, and started whole new fields of investigations.

Apart from his penetrating work in Mathematics and the Sciences, he devoted much of his time to improving the teaching of mathematics in secondary schools in the Soviet Union, and in providing special schools for the mathematically gifted - which were very successful. Famous are also his efforts to capture in quantitative form some aspects of Russian poetry, especially that of Pushkin. It is told that "it was fascinating to hear him lecture on this, whether one understood Russian or not." In 1942 Kolmogorov married Anna Dmitriyevna Egorov. He did not have children of his own.

While being acknowledged without question in science, Kolmogorov was also blessed with social recognition. The USSR conferred to him seven orders of Lenin, the Order of the October Revolution, and also the high title of Hero of Socialist Labour; he gained Lenin prizes and State prizes. He occupies the first place among all Soviet mathematicians in the number of foreign academies and scientific societies that have elected him as member. These number over twenty, among them

- the Royal Netherlands Academy of Sciences (1963),
- the London Royal Society (1964),
- the USA National Society (1967),
- the Paris Academy of Sciences (1968),
- the Polish Academy of Sciences,
- the Rumanian Academy of Sciences (1956),
- the German Academy of Sciences Leopoldina (1959),
- the American Academy of Sciences and Arts in Boston (1959).

He was presented with honorary doctorates from the universities of Paris, Berlin, Warsaw, Stockholm, etc. He was elected a honorary member of the Moscow, London, Indian, and Calcutta Mathematical Societies, of the London Royal Statistical Society, the International Statistical Institute, and the American Meteorological Society. In 1963 he was awarded the International Bolzano prize.

<https://mathshistory.st-andrews.ac.uk/Biographies/Euler/>

Post-publication activity

Curator: [Paul M.B. Vitanyi](#)

Leonhard Euler

Quick Info

Born

15 April 1707
[Basel, Switzerland](#)

Died

18 September 1783
St Petersburg, Russia



[View seventeen larger pictures](#)

Summary

Leonhard Euler was a Swiss mathematician who made enormous contributions to a wide range of mathematics and physics including analytic geometry, trigonometry, geometry, calculus and number theory.

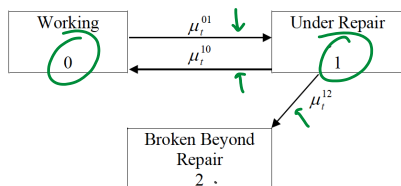
Biography

Leonhard Euler's father was Paul Euler. Paul Euler had studied theology at the University of Basel and had attended [Jacob Bernoulli's](#) lectures there. In fact Paul Euler and [Johann Bernoulli](#) had both lived in [Jacob Bernoulli's](#) house while undergraduates at Basel. Paul Euler became a Protestant minister and married Margaret Brucker, the daughter of another Protestant minister. Their son Leonhard Euler was born in Basel, but the family moved to Riehen when he was one year old and it was in Riehen, not far from Basel, that Leonard was brought up. Paul Euler had, as we have mentioned, some mathematical training and he was able to teach his son elementary mathematics along with other subjects.

Leonhard was sent to school in Basel and during this time he lived with his grandmother on his mother's side. This school was a rather poor one, by all accounts, and Euler learnt no mathematics at all from the school. However his interest in mathematics had certainly been sparked by his father's teaching, and he read mathematics texts on his own and took some private lessons. Euler's father wanted his son to follow him into the church and sent him to the University of Basel to prepare for the ministry. He entered the University in 1720, at the age of 14, first to obtain a general education before going on to more advanced studies. [Johann Bernoulli](#) soon discovered Euler's great potential for mathematics in private tuition that Euler himself engineered. Euler's own account given in his unpublished autobiographical writings, see [\[1\]](#), is as follows:-

Question 7 ALTAM WA

(7 points) An actuarial student in your department is using the following 3-state Markov model to price a one-year consumer warranty for a television.



state j $\frac{d}{dt} {}_tP_0^{00}$ state 0 (+) (-)
 $\frac{d}{dt} {}_tP_0^{01}$ state 1 (+) (-) (-)

The product has the following features:

- (i) The warranty is sold at the time that a consumer purchases the television and pays a replacement cost of 1000 at the end of the half-year during which the television becomes Broken Beyond Repair.
- (ii) Premiums are payable at the beginning of each half-year, but are waived if the television is not Working at the time a premium is due.
- (iii) All televisions are in the Working state at time of purchase.

- (a) (2 points) Write down the Kolmogorov forward differential equations with associated boundary conditions (initial conditions) for ${}_tP_0^{00}$, ${}_tP_0^{01}$, and ${}_tP_0^{02}$ under this model.

$$\frac{d}{dt} {}_tP_0^{00} = \sum_{\substack{k=0 \\ k \neq j=0 \\ k=1,2}}^2 ({}_tP_0^{0k} \mu_t^{k0} - {}_tP_0^{00} \mu_t^{0k}) = ({}_tP_0^{01} \mu_t^{10} - {}_tP_0^{00} \mu_t^{01}) + \cancel{({}_tP_0^{02} \mu_t^{20} - {}_tP_0^{00} \mu_t^{02})}$$

$$\frac{d}{dt} {}_tP_0^{00} = {}_tP_0^{01} \mu_t^{10} - {}_tP_0^{00} \mu_t^{01}$$

$$\frac{d}{dt} {}_tP_0^{01} = \sum_{\substack{k=0 \\ k \neq 1 \\ k=0,2}}^2 ({}_tP_0^{0k} \mu_t^{k1} - {}_tP_0^{01} \mu_t^{1k}) = ({}_tP_0^{00} \mu_t^{01} - {}_tP_0^{01} \mu_t^{10}) + \cancel{({}_tP_0^{02} \mu_t^{21} - {}_tP_0^{01} \mu_t^{12})}$$

$$\frac{d}{dt} {}_tP_0^{01} = {}_tP_0^{00} \mu_t^{01} - {}_tP_0^{01} \mu_t^{10} - {}_tP_0^{01} \mu_t^{12} \quad *$$

$$\frac{d}{dt} {}_tP_0^{02} = \sum_{\substack{k=0 \\ k \neq 2 \\ k=0,1}}^2 ({}_tP_0^{0k} \mu_t^{k2} - {}_tP_0^{02} \mu_t^{2k}) = \cancel{({}_tP_0^{00} \mu_t^{02} - {}_tP_0^{02} \mu_t^{20})} + \cancel{({}_tP_0^{01} \mu_t^{12} - {}_tP_0^{02} \mu_t^{21})}$$

$$\frac{d}{dt} {}_tP_0^{02} = {}_tP_0^{01} \mu_t^{12}$$

$${}_0P_0^{00} = 1 \quad {}_0P_0^{01} = 0 \quad {}_0P_0^{02} = 0$$

If the candidate omits

$\frac{d}{dt}$

no credit at all

The forces of transition in this model for a television purchased t years ago are:

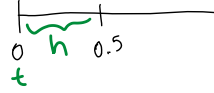
$$\mu_t^{01} = 0.5 + 0.6t$$

$$\mu_t^{10} = 0.2$$

$$\mu_t^{12} = 2t$$

The student has applied the Forward Euler approximation to the Kolmogorov forward differential equations with a step size of $h = 0.5$ to calculate probabilities for this model. You have verified that, using this approach, ${}_{0.5}p_0^{00} = 0.75$.

(b) (2 points) Using the student's approach:



(i) Calculate ${}_{0.5}p_0^{01}$.

(ii) Show that the probability that a television will become Broken Beyond Repair within a year of purchase is 0.18 to the nearest 0.01. You should calculate the probability to the nearest 0.001.

$$\begin{aligned}
 (i) \quad 0 + 0.5 p_0^{01} &= \cancel{0 p_0^{01}} + 0.5 \left(\underset{\uparrow}{0 p_0^{00}} \mu_0^{01} - \cancel{0 p_0^{01}} \mu_0^{10} - \cancel{0 p_0^{10}} \mu_0^{12} \right) \\
 &= (0.5)(1)(0.5) = \boxed{0.25} \star
 \end{aligned}$$